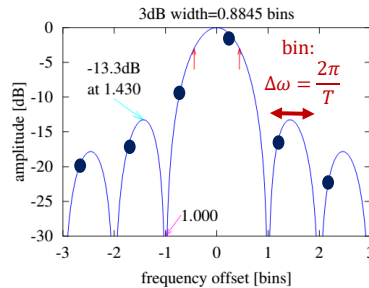


Window functions and their properties

Further on we analyze the properties of various window functions with similar figures, as the right one for the rectangular window. The DFT points are positioned equidistantly with one point / bin. Note the logarithmic vertical scale!

Which is the best window function? It depends on the goal of our analysis, do we want to have **good frequency resolution**, **small spectral leakage**, or **small amplitude error**?

Good frequency resolution is obtained if the central peak of $f_W(\omega)$ is as narrow as possible. This can be characterized by the distance between the 3dB suppression points, i.e. the frequencies where the power is halved ($10 \cdot \log(1/2) = -3.0102...$)



If we want **spectral leakage** to be **small**, we need a window function whose Fourier transform has the side peaks suppressed as much as possible. This can be characterized by the suppression of the first side peak ($-13.2\text{dB} \rightarrow 4.7\%$)

If we want to **accurately determine the amplitude of our signal** with ω_0 circ. frequency, we need to have a very flat $f_W(\omega)$ within the interval of $2\pi/T$ around the central frequency. The worst case is, when ω_0 is halfway between two discrete frequency points. Even in this case we can accurately read the signal amplitude if the window function is flat in the frequency space. In this case, however, our frequency resolution will be poor by definition.

Note: it is useful to normalize the window functions!

$$F(t) = A e^{i\omega_0 t} \Rightarrow f_W(\omega_0) = A \cdot w(0), \quad w(0) = \frac{T/2}{\int_{-T/2}^{T/2} W(t) dt}$$

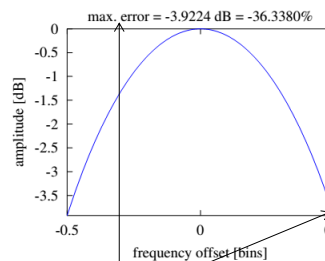
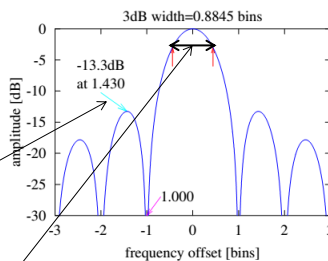
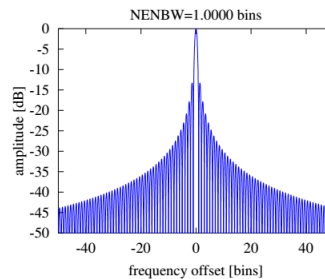
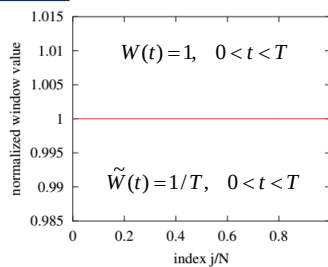
Normalized window function:
$$\tilde{W}(t) = \frac{W(t)}{\int_{-T/2}^{T/2} W(t) dt}$$

Taking advantage of this, $f_{\tilde{W}}(\omega_0) = A$ is valid regardless of the choice of the window function, i.e. the signal amplitude can be easily read from the height of the central peak.

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Rectangular window

Rectangular



The first side peak suppression: -13.3dB (4.7%). This is a **strong spectral leakage**.

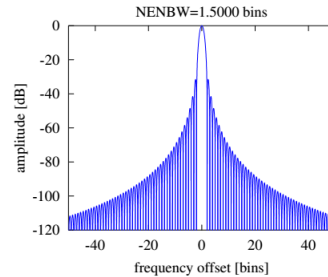
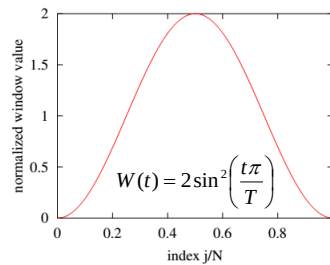
Frequency resolution (-3dB point spacing): 0.8845 bin. This is the **best frequency resolution** for a certain T , the other window functions have worse frequency resolution!

If ω_0 is halfway between two discrete frequencies, we measure 36% less amplitude than if ω_0 coincides with a discrete frequency. This is very **poor amplitude accuracy**.

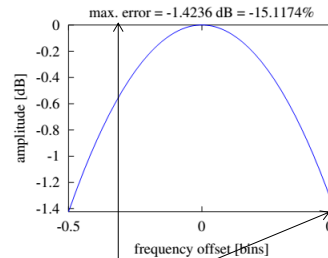
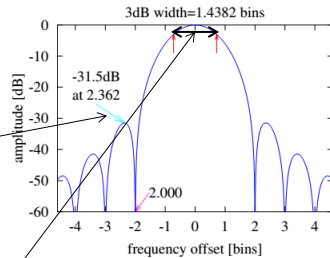
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Hanning window

Hanning



The first side peak suppression: -31.5dB (0.07%). Relatively **low spectral leakage**, significantly **better than the rectangular window**.



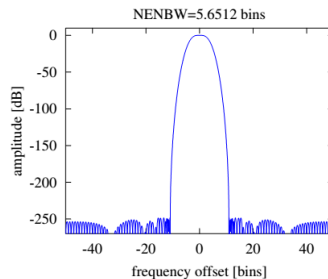
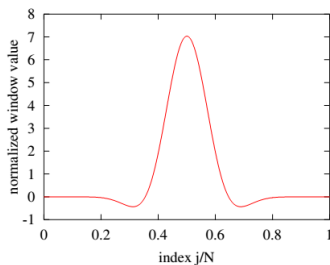
Frequency resolution (-3dB point spacing): 1.44 bin. This is worse than the rectangular window, but not too bad.

If ω_0 is halfway between two discrete frequencies, we measure 15% less amplitude than if ω_0 coincides with a discrete frequency. This is still a **poor amplitude accuracy**.

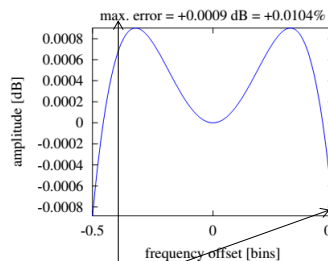
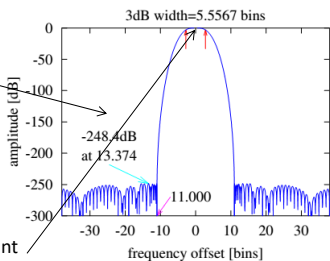
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Flattop window

HFT248D



The first side peak suppression: -248dB. (1.6×10^{-25}) **Very low spectral leakage!**



Frequency resolution (-3dB point spacing): 5.5 bin. This is **very poor** frequency resolution! This is not a surprise, as the effective measurement time with nonzero window is just a fraction of T.

Wherever ω_0 is located between two discrete frequencies, the same value is obtained for the amplitude to within **0.01%**. The **flattop window is commonly used when good amplitude accuracy is the goal** and frequency resolution is less important!

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